DATA ANALYTICS PROJECT REPORT

Retail Sales Analysis

A business review of sales, cost, profit, category performance and city-level results.

TOOLS

Python • Pandas • Matplotlib

DATA PERIOD

January–March 2026

Portfolio practice project · Prepared October 2026

Executive takeaway
The dataset records PKR 117,300 in sales and PKR 24,430 in calculated profit across 18 orders. Furniture contributes the most total profit, while Electronics has the highest profit margin. These are descriptive findings from a small practice dataset, not evidence of a real retailer’s performance.

1. Executive Summary

This project analyzes a small retail sales dataset covering January to March 2026. Using Python, Pandas and Matplotlib, it calculates sales, cost, profit and profit margin, then compares performance by category, month, city and product. The aim is to demonstrate a clear, repeatable beginner-level data-analysis workflow.

TOTAL SALES	TOTAL COST	TOTAL PROFIT
PKR 117,300	**PKR 92,870**	**PKR 24,430**
PROFIT MARGIN	ORDERS	UNITS SOLD
20.83%	**18**	**73**

Key findings

Finding	Result	Interpretation
Highest-sales city	Lahore — PKR 49,200	Largest sales total in this sample.
Highest-sales month	February — PKR 45,650	Sales rose from January, then eased in March.
Highest-profit category	Furniture — PKR 13,300	Largest total profit contribution.
Highest-margin category	Electronics — 26.01%	Highest profit per rupee of sales in the sample.

All amounts are in Pakistani rupees (PKR). Sales = Quantity × Unit Price; cost = Quantity × Unit Cost; profit = Sales – Cost.

2. Data & Methodology

Dataset overview

Item	Description
Time period	1 January–26 March 2026
Records	18 rows and 18 unique order IDs in this version
Fields	Order date, city, category, product, quantity, unit price and unit cost
Tools	Python, Pandas for inspection and aggregation; Matplotlib for charts

Calculated metrics

Metric	Calculation
Sales	Quantity × Unit_Price_PKR
Total cost	Quantity × Unit_Cost_PKR
Profit	Sales – Total cost
Profit margin	Profit ÷ Sales × 100
Average order value	Total sales ÷ distinct orders = PKR 6,516.67

Workflow

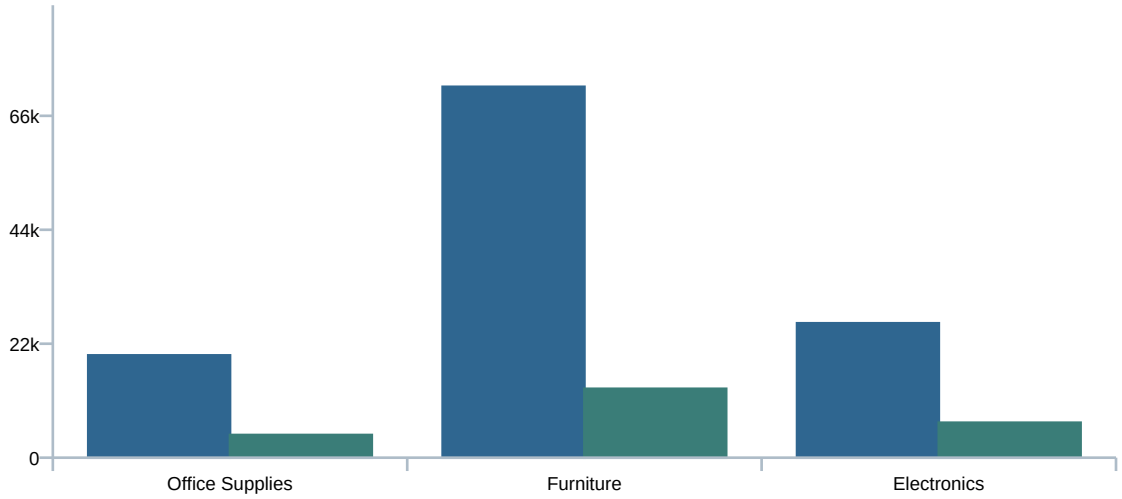
1. Load the CSV and inspect columns, data types and structure.
2. Check missing values, duplicates, invalid dates and numeric values.
3. Create sales, cost and profit columns.
4. Aggregate results by category, month, city and product.
5. Visualize comparisons and trends, then write plain-language findings.

Quality-control note: the figures in this report match the previously verified calculations for the supplied 18-row practice dataset. If the CSV changes, rerun the notebook and update all figures.

3. Category Performance

Furniture generated the largest sales and profit totals, while Electronics had the strongest profit margin. The category with the highest total profit is not necessarily the category with the highest margin.

Sales vs. profit by category

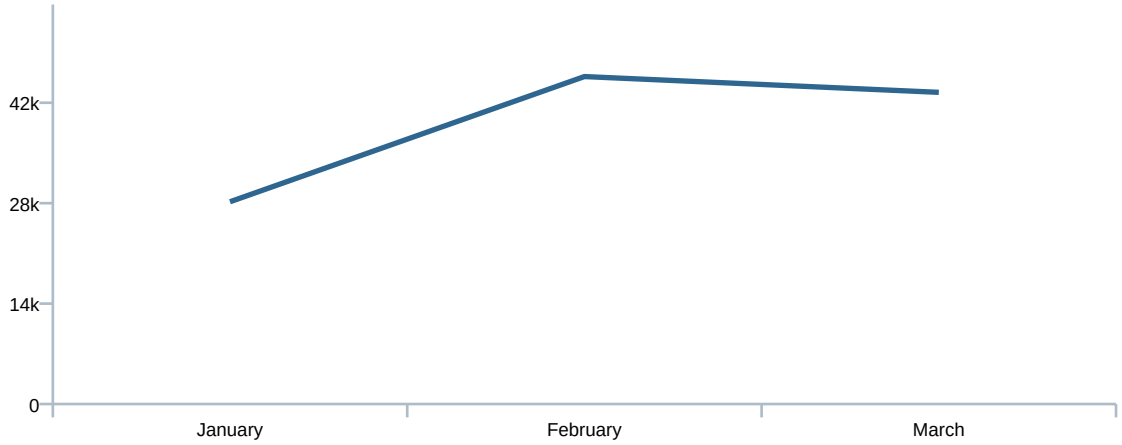


Category	Sales (PKR)	Profit (PKR)	Margin
Office Supplies	19,750	4,380	22.18%
Furniture	71,600	13,300	18.58%
Electronics	25,950	6,750	26.01%

Interpretation: Furniture is the largest contributor to total profit. Electronics retains the largest share of sales as profit in this sample. The dataset is too small to conclude these patterns will persist.

4. Monthly Sales Trend

Monthly sales (PKR)

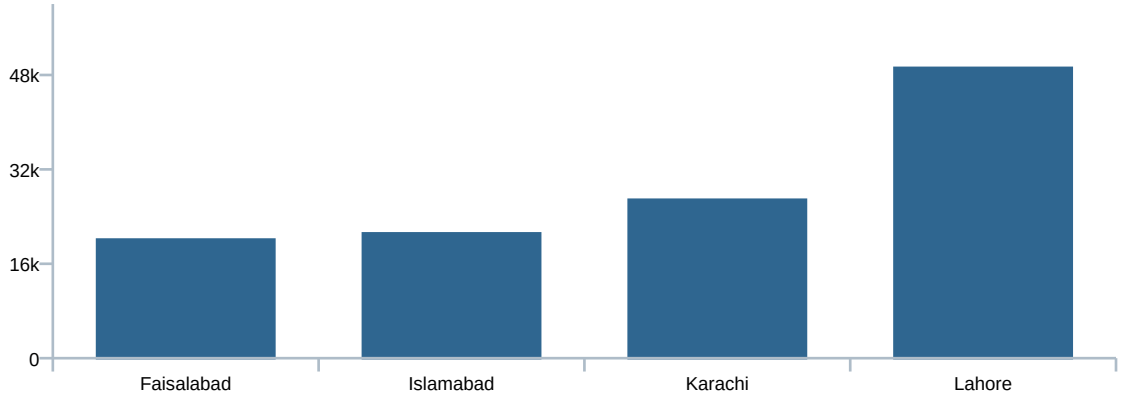


Month	Sales (PKR)	Change vs. prior month
January	28,200	—
February	45,650	+17,450 (+61.9%)
March	43,450	–2,200 (–4.8%)

February was the strongest month. Sales rose approximately 61.9% from January to February, then fell approximately 4.8% in March. March remained above January. With only three months of data, this is a short-period comparison, not a reliable seasonal forecast.

5. City Performance

Sales by city (PKR)



City	Sales (PKR)	Profit (PKR)	Margin
Faisalabad	20,100	5,000	24.88%
Islamabad	21,150	4,900	23.17%
Karachi	26,850	5,460	20.34%
Lahore	49,200	9,070	18.44%

Lahore leads in sales and total profit, while Faisalabad has the highest city-level margin. This illustrates why both totals and ratios should be compared.

6. Product-Level Results

Product	Units sold	Sales (PKR)	Profit (PKR)
Office Chair	4	50,000	8,800
Wireless Mouse	9	16,200	4,050
Desk Lamp	9	21,600	4,500
USB-C Cable	15	9,750	2,700
Printer Paper	13	11,700	2,080
Notebook Pack	23	8,050	2,300

Office Chair produced the highest sales total (PKR 50,000) and highest total profit (PKR 8,800), despite only four units sold. Notebook Pack had the highest unit volume (23), but its lower price meant its sales total was much smaller. Unit volume and revenue should be evaluated separately.

7. Business Recommendations

1. Review Furniture economics

Furniture drives the largest sales and profit totals but has the lowest category margin (18.58%). Review supplier costs, discounts and product mix before making pricing changes.

2. Investigate Electronics margin

Electronics has the highest observed category margin (26.01%). Validate the result across more months and products before expanding investment.

3. Study Lahore’s sales mix

Examine which products and order types drive Lahore’s sales and whether its lower margin is consistent in a larger dataset.

4. Track results over time

Collect more months of data and compare against targets. Three months are insufficient to establish seasonality or forecast demand reliably.

5. Improve dataset realism

If available, add returns, discounts, shipping, taxes, inventory, customer segments and transaction status to better reflect operating performance.

8. Limitations & Assumptions

• The dataset is a small, fictional/practice dataset with 18 rows and less than three full months. It is not evidence about an actual retailer.

• Each row is treated as one order line. In this version each row has a unique Order_ID; if future data contains multiple lines per order, AOV must use distinct order IDs rather than row count.

• Profit is sales minus the supplied unit cost. It is not net profit because overhead, wages, shipping, taxes, returns and other operating expenses are excluded.

• Customer details, marketing spend, inventory availability, discounts and returns are not provided.

• Findings are descriptive: they summarize this sample but do not establish causes or support robust forecasting.

• Results may change with more observations or a different time period.

9. Conclusion

The analysis demonstrates an introductory workflow covering data inspection, validation, metric creation, aggregation, visualization and business interpretation. The sample records PKR 117,300 in sales, PKR 24,430 in calculated profit and a 20.83% overall profit margin. Furniture contributes the highest total profit, Electronics has the highest category margin, Lahore leads city sales, and February is the strongest month.

Recommended next step: publish the notebook, source CSV, this report and a README together in an organized portfolio repository. Clearly label the dataset as practice data and be ready to explain formulas, charts, limitations and recommendations.

Appendix: Key formulas

Metric	Formula
Sales	Quantity × Unit Price
Cost	Quantity × Unit Cost
Profit	Sales – Cost
Profit margin	Profit ÷ Sales × 100
Average order value	Total sales ÷ distinct orders

Prepared as an educational portfolio sample. Revalidate values if the underlying dataset changes.